



University of Moratuwa

Department of Electronic and Telecommunication Engineering

Faculty of Engineering

Class D Audio Amplifier

Design, Simulation and Practical Implementation

Module: EN 2111 - Electronic Circuit Design

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Introduction

1.1 Background and Motivation

Class-D amplifiers are widely used in modern audio systems because of their high efficiency. Unlike conventional linear amplifiers, they amplify audio by converting it into a high-frequency Pulse Width Modulated (PWM) signal, allowing the output MOSFETs to operate as switches. This greatly reduces power loss while maintaining good audio quality. The aim of this project was to design, simulate, and implement a Class-D audio amplifier using commonly available components. Several design iterations were carried out to improve the PWM generation, gate driving, and output filtering stages before achieving the final design.

1.2 Objectives

The objectives of this project are to:

- Design and implement a complete Class-D audio amplifier.
- Generate PWM using a high-frequency triangular carrier and comparator.
- Design the MOSFET gate driver and output stage.
- Recover the amplified audio signal using an LC low-pass filter.
- Verify the design through simulation and practical testing.

1.3 Scope of the Report

This report covers the design, simulation, implementation, and testing of the Class-D audio amplifier, including the pre-amplifier, PWM generator, gate driver, output stage, LC filter, and PCB design.

1.4 Report Organisation

Chapter 2 presents the design and theory of each subsystem, while the remaining chapters cover simulation, implementation, testing, PCB design, results, challenges, and the final conclusions.

Theoretical Background

2.1 Pre-Amplifier

2.1.1 Purpose and Role in the Signal Chain

The pre-amplifier stage serves as the front-end interface between audio sources such as smartphones, media players or DACs and the PWM stage of the Class-D Amplifier. The standard line-level audio equipment delivers audio signals ranging from 200 millivolts to 1 volts. The comparator LM311D requires an audio signal which should closely match the amplitude of the triangular wave to create a good PWM with high modulation index. For this purpose a pre-amplifier is needed. By maximizing the modulation index, the SNR increases and the voltage utilization factor of the output power rails also increases.

Beyond this, The pre-amplifier stage fullfills these roles:

- **Impedance Buffering:** Presenting a high input impedance to the incoming source to eliminate signal loading, ensuring flat frequency transmission from the playback device.
- **DC Isolation and Biasing:** Utilizing an AC-coupling network to strip away any residual DC offsets from the input source, ensuring that the audio signal centers precisely around the analog ground (0 V) of the split-supply system.

2.1.2 Design Considerations and Circuit Analysis

The complete pre-amplifier is designed around the **NE5532** dual operational amplifier which has $5 \text{ nV}/\sqrt{\text{Hz}}$ noise density and 10 MHz small-signal bandwidth which makes it perfect for our application.

1. Input Gain Stage

The first stage (U_1) is configured as an **inverting amplifier**. The mid-band voltage gain (A_{v1}):

$$A_{v1} = -\frac{R_2}{R_1} = -\frac{33 \text{ k}\Omega}{2.2 \text{ k}\Omega} = -15 \quad (\approx 23.5 \text{ dB}) \quad (2.1)$$

The lower -3 dB cut-off frequency (f_{L1}) of this initial block:

$$f_{L1} = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi \times 2.2 \text{ k}\Omega \times 10 \mu\text{F}} \approx 7.23 \text{ Hz} \quad (2.2)$$

This pole is placed below the nominal 20 Hz audible threshold to avoid phase shifts in the bass range. Noise suppression is managed by the feedback capacitor C_2 in parallel with R_2 forming a low-pass filter with a corner frequency (f_{H1}):

$$f_{H1} = \frac{1}{2\pi R_2 C_2} = \frac{1}{2\pi \times 33 \text{ k}\Omega \times 47 \text{ pF}} \approx 102.6 \text{ kHz} \quad (2.3)$$

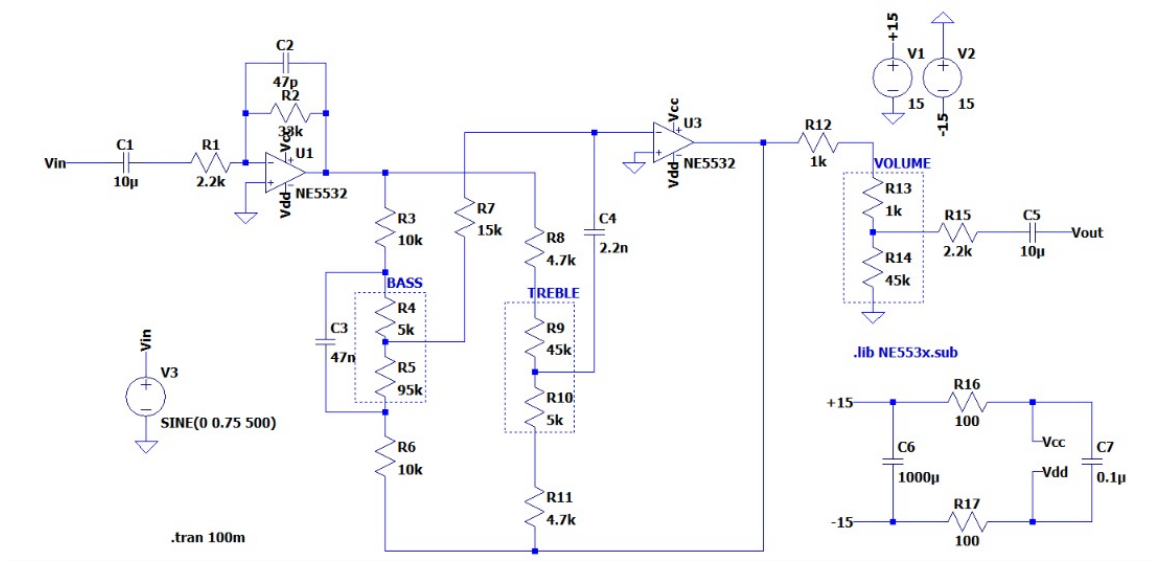
2. Active Baxandall Tone Control Stage

The output of U_1 feeds into an active **Baxandall tone control** network built around the second operational amplifier stage (U_3).

- **Bass Control Loop:** Resistors R_3 , R_6 , and capacitor C_3 (47 nF) function together with the BASS linear potentiometer ($R_4 + R_5 = 100 \text{ k}\Omega$). At low frequencies, C_3 behaves as an open circuit, allowing the position of the wiper to adjust the relative feedback resistance ratio, boosting or cutting low frequencies. At high frequencies, C_3 shorts out the potentiometer, rendering the bass control inactive.
- **Treble Control Loop:** Resistors R_7 , R_8 , R_{11} , and capacitor C_4 (2.2 nF) form the treble adjustment route across the TREBLE potentiometer ($R_9 + R_{10} = 50 \text{ k}\Omega$). Because C_4 features low impedance at high frequencies, it allows the treble potentiometer to modify the high-frequency feedback path, controlling high-frequency shelving adjustments without affecting the lower audio register.

3. Volume Control and Output Coupling

The output of the tone equalizer stage (U_3) passes through a isolation resistor $R_{12} = 1 \text{ k}\Omega$ to a variable passive potential divider labeled VOLUME ($R_{13} = 1 \text{ k}\Omega$, $R_{14} = 45 \text{ k}\Omega$). This potentiometer scales the processed signal down to the ideal peak matching envelope of the triangular wave to generate PWM signal with maximum modulation index. Finally, an impedance-matching resistor $R_{15} = 2.2 \text{ k}\Omega$ and an output AC-coupling capacitor $C_5 = 10 \mu\text{F}$ strip away any accumulated operational amplifier DC offsets before routing the clean baseband audio into the *LM311D* PWM comparator.

Figure 2.1: Pre-Amplifier Circuit on *LTSpice*

2.1.3 Pre-Amplifier Simulation

The Pre amplifier circuit simulated in *LTSpice*.

1. AC Small-Signal Sweep (Bode Analysis)

To evaluate the control envelope of the active Baxandall network, an AC small-signal sweep was simulated from 1 Hz to 100 kHz with a 1 V reference AC excitation input. The potentiometer components were modeled under variable resistance ratios to inspect the system response across distinct user settings:

- **Flat Audio Passband:** With both the BASS (R_4, R_5) and TREBLE (R_9, R_{10}) potentiometer networks configured to their nominal midpoints, the circuit demonstrates a highly linear, flat magnitude transmission profile across the audible core range.

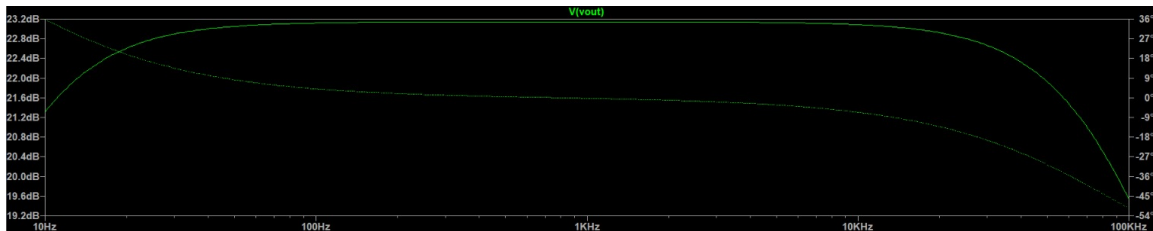


Figure 2.2: Flat Response

- **Low-Frequency Adjustment (Bass Boost/Cut):** Adjusting the BASS potentiometer wiper sweeps the low-frequency shelf characteristic below 200 Hz. At maximum boost configuration, an approximate +12 dB shelf gain enhancement is ob-

served near the lower frequencies, whereas max cut introduces corresponding low-end attenuation without distorting mid-range frequencies.

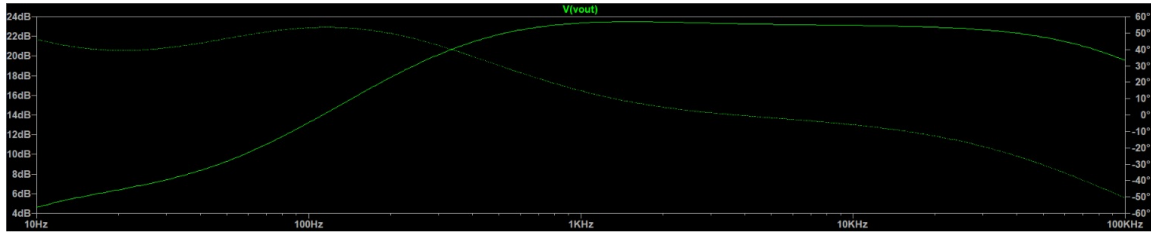


Figure 2.3: Bass Boost

2. Total Harmonic Distortion (THD) Analysis

Because the processed audio signal directly governs the pulse-width generation thresholds inside the downstream *LM311D* comparator, maintaining minimal harmonic coloration is vital. A high-resolution discrete Fourier Transform (DFT) analysis was performed using a 500 Hz sinusoidal test tone with a representative input amplitude of 750 mV_{peak} as modeled by source V_3 .

Fourier components of V(Vout)
N-Period=1
DC component:0.0133062

Harmonic Number	Frequency [Hz]	Fourier Component	Normalized Component	Phase [degree]
1	5.000e+02	1.022e+01	1.000e+00	48.29°
2	1.000e+03	1.040e-03	1.018e-04	0.82°
3	1.500e+03	5.302e-03	5.189e-04	-74.26°
4	2.000e+03	9.987e-04	9.775e-05	-23.88°
5	2.500e+03	4.059e-03	3.973e-04	99.29°
6	3.000e+03	9.345e-04	9.146e-05	-5.17°
7	3.500e+03	4.623e-03	4.524e-04	-50.61°
8	4.000e+03	8.686e-04	8.501e-05	-33.05°
9	4.500e+03	2.250e-03	2.202e-04	123.06°

Partial Harmonic Distortion: 0.084605%
Total Harmonic Distortion: 0.243278%

Figure 2.4: Total Harmonic Distortion Analysis

The calculated THD value falls well within ideal parameters. This exceptionally low distortion value confirms that the *NE5532* operational amplifiers preserve fundamental signal composition with zero clipping or crossover limitations before the audio signal enters the high-speed PWM conversion stage.

2.2 PWM Generation

The Class D amplifier uses pulse width modulation to convert the analog audio signal into a high-frequency switching waveform. In this method, the audio signal is compared

with a high-frequency triangular carrier. The output of the comparator becomes a PWM signal whose duty cycle changes according to the instantaneous amplitude of the audio signal. This PWM signal is then sent to the gate driver and MOSFET output stage.

2.2.1 Triangular Wave Generator

The triangular wave generator provides the carrier signal for PWM generation. Since the audio input is limited to approximately 20 Hz – 20 kHz , the carrier frequency must be significantly higher. A target frequency of approximately 350 kHz was selected. Initially, an NE555 timer-based circuit was designed and tested, using an RC network to produce an approximate triangular waveform. This approach was chosen due to the simplicity, low cost, and ease of configuration of the NE555 as an oscillator.

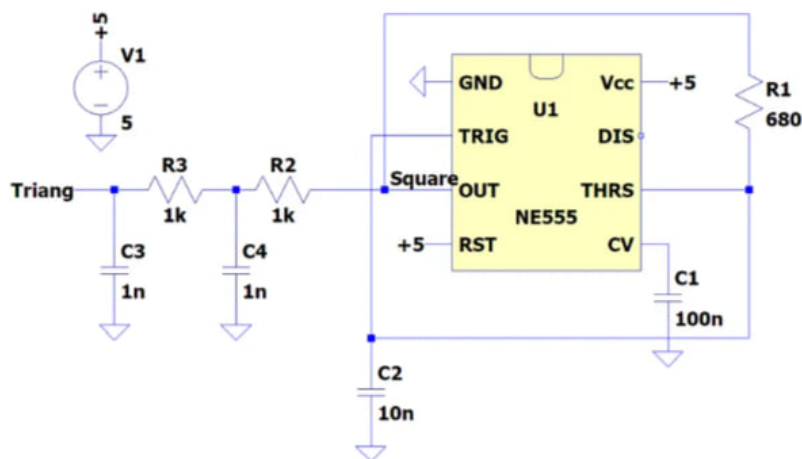


Figure 2.5: Initial NE555-based triangular wave generation circuit

The NE555-based design produced a triangular waveform with a significant DC offset because its output switches between 0 V and V_{CC} , resulting in a carrier centred above 0 V . This shifted the comparator threshold and caused incorrect PWM duty cycles even when the audio input was zero. In addition, the RC charging and discharging process is exponential rather than linear, leading to a non-ideal triangular waveform and potential distortion. Due to these limitations, the design was replaced with a high-speed AD8065 op-amp implementation.

The final design uses two AD8065 stages: a relaxation oscillator (Schmitt trigger) to generate a square wave, followed by an inverting integrator that converts the square wave into a stable, low-distortion triangular waveform suitable for PWM generation.

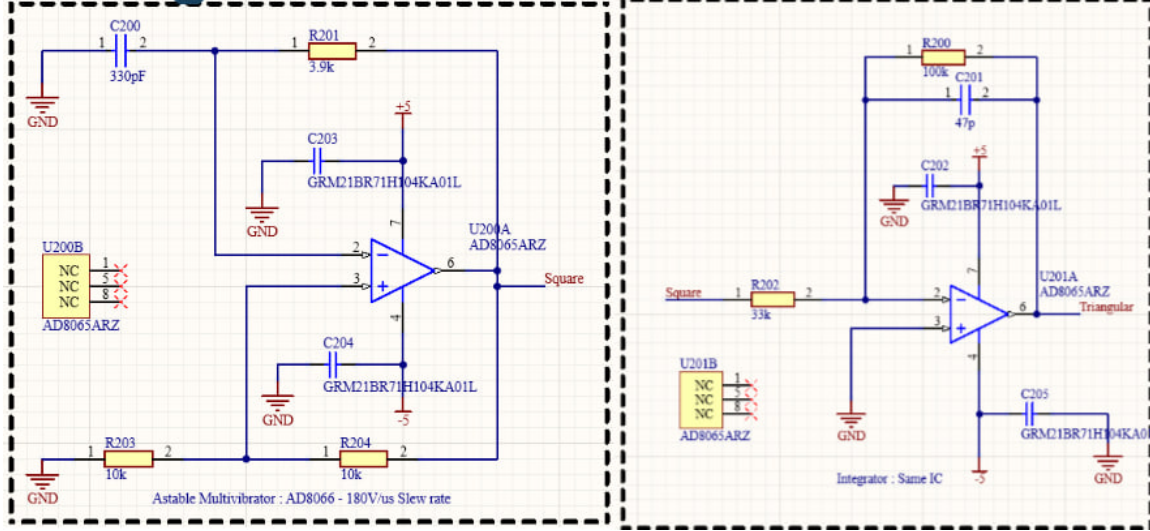


Figure 2.6: Final AD8065-based triangular wave generator using oscillator and integrator stages

The final triangular wave generator was implemented using the AD8065 high-speed op-amp, selected for its high bandwidth and slew rate, which are necessary for operation at the target carrier frequency of approximately $350, kHz$. The circuit consists of a relaxation oscillator that generates a square wave and an inverting integrator that converts the square wave into a triangular waveform.

The oscillator frequency is given by

$$f = \frac{1}{2RC \ln \left(\frac{1+\beta}{1-\beta} \right)} \quad (2.4)$$

where $\beta = \frac{R_{lower}}{R_{upper} + R_{lower}}$. Using $R = 3.9, k\Omega$, $C = 330, pF$, and equal feedback resistors of $10, k\Omega$ ($\beta = 0.5$), the calculated oscillation frequency is approximately $354, kHz$, which closely matches the design target.

The square wave output is applied to an inverting integrator with $R_i = 33, k\Omega$ and $C_i = 47, pF$. The integrator produces a triangular waveform whose amplitude depends on the carrier frequency and the integrator time constant. A $100, k\Omega$ resistor was connected in parallel with the integrator capacitor to provide a DC feedback path and prevent output saturation caused by op-amp offset voltages and bias currents.

The measured waveform closely followed the simulated result, although some noise and distortion were observed. These effects are mainly attributed to parasitic capacitances, supply noise, PCB layout limitations, square-wave feedthrough, and oscilloscope probe loading. Despite these non-idealities, the circuit produced a stable triangular carrier suitable for PWM generation in the Class-D amplifier.

2.2.2 Comparator Stage

The comparator stage generates the PWM signal by comparing the amplified audio signal with the high-frequency triangular carrier. When the audio signal is greater than the carrier, the comparator output is HIGH; otherwise, it is LOW. As the audio amplitude varies, the duty cycle of the output changes proportionally, producing the PWM waveform required for Class-D amplification.

For a symmetric triangular carrier, the PWM duty cycle can be approximated by

$$D(t) = \frac{1}{2} + \frac{V_{audio}(t)}{2V_{tri,pk}} \quad (2.5)$$

To maintain linear PWM operation, the audio signal amplitude must remain smaller than the carrier amplitude. If the audio signal exceeds the carrier amplitude, overmodulation occurs, resulting in clipping and increased distortion.

During the mid-evaluation stage, PWM generation was implemented using the LM393 comparator. Although it successfully demonstrated PWM operation, its open-collector output required a pull-up resistor, resulting in slower rise times and reduced switching performance at high carrier frequencies.

For the final design, the LM393 was replaced with the LM311D comparator, which offers faster switching characteristics and cleaner output transitions. This change improved PWM signal quality and reduced timing errors at the selected carrier frequency of approximately $350, kHz$.

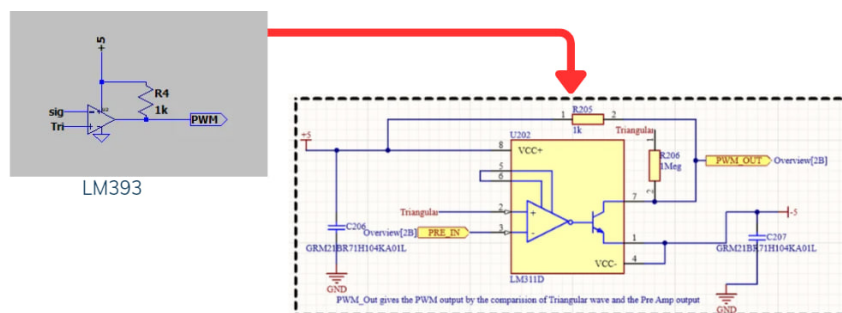


Figure 2.7: Transition from the LM393-based PWM generator used during mid-evaluation to the final LM311D-based high-frequency PWM generation circuit

In the final implementation, the audio signal from the pre-amplifier and tone control stage is applied to one input of the LM311D, while the triangular carrier generated by the AD8065 oscillator-integrator circuit is applied to the other. The resulting PWM output is then fed to the MOSFET gate driver stage.

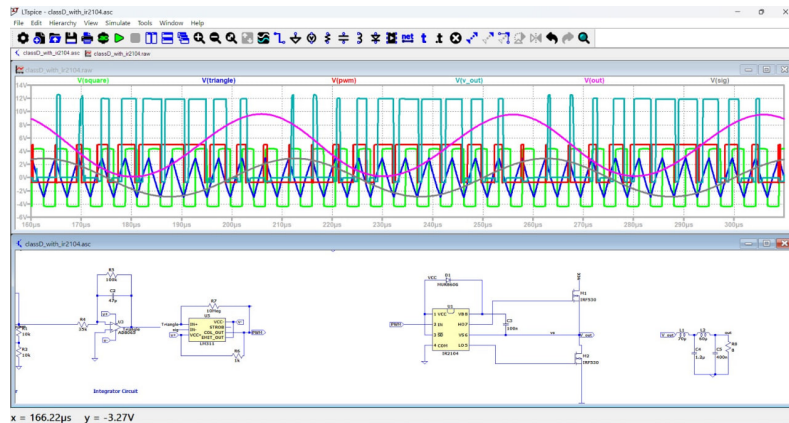


Figure 2.8: Simulation of PWM generation using the LM311D comparator

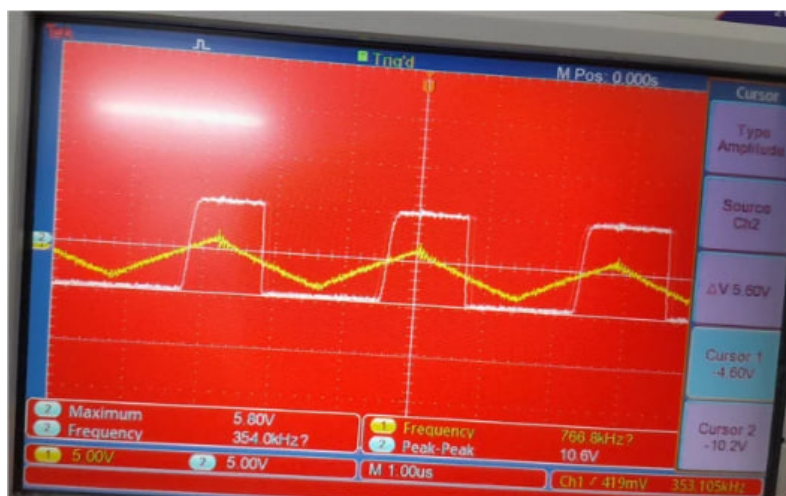


Figure 2.9: Measured high-frequency PWM output waveform generated using the LM311D comparator

The measured PWM waveform closely matched the simulation results. Minor variations were observed due to noise, PCB parasitics, and supply fluctuations. Since the PWM signal directly controls the MOSFET switching stage, proper decoupling, grounding, and carrier waveform quality are important to minimize jitter and distortion.

2.2.3 PWM Frequency Selection

The PWM switching frequency was set to approximately 350 kHz , much higher than the audio band ($\sim 20\text{ kHz}$), allowing effective filtering of switching components while preserving the audio signal. The ratio $f_s/f_{\text{audio,max}} \approx 17.5$ provides adequate separation for proper reconstruction.

Although higher switching frequency improves filtering and reduces ripple, it increases switching losses and driver demands. Therefore, 350 kHz was chosen as a practical trade-

off between efficiency, signal quality, and hardware limitations, and is compatible with the overall system design.

2.3 MOSFET Gate Driver and Output Stage

2.3.1 MOSFET Selection Criteria

Candidates for mosfets found locally in Sri Lanka:

- IRF530
- IRFZ44N

Table 2.1: MOSFET Comparison: IRFZ44N vs IRF530 for Half-Bridge Output Stage

Parameter	IRFZ44N	IRF530
V_{DS} (max)	55 V	100 V
I_D (max, $T_C = 25^\circ\text{C}$)	49 A	14 A
$R_{DS(on)}$ (max)	17.5 m Ω	160 m Ω
Q_g (total gate charge)	63 nC	26 nC
C_{iss} (input capacitance)	1470 pF	670 pF
t_r (rise time)	60 ns	34 ns
t_f (fall time)	45 ns	24 ns
$V_{GS(th)}$ (threshold)	2–4 V	2–4 V
P_D (max dissipation)	94 W	74 W
300 kHz Performance Higher conduction efficiency Lower switching loss		

At 300 kHz, gate charge and capacitance dominate switching performance. IRF530 has lower Q_g and C_{iss} , giving lower switching loss and better high-frequency operation. IRFZ44N has lower $R_{DS(on)}$ but higher gate-drive loss. Hence, IRF530 was selected for the final design.

$\therefore$ Mosfet used for the half bridge output stage $\rightarrow$ IRF530.

Calculating source and sink current required to turn on mosfets

Source current:

$$I_{source} = \frac{Q_g}{t_r} \quad (2.6)$$

Sink current:

$$I_{sink} = \frac{Q_g}{t_f} \quad (2.7)$$

$$T = \frac{1}{300kHz} \quad (2.8)$$

At a switching frequency of 300 kHz, the switching period is $3.33 \mu s$, corresponding to a half-period of $1.67 \mu s$. A common design guideline is to complete the MOSFET switching transition within 5% of the half-period, corresponding to approximately 83 ns. This minimizes switching losses and distortion while maintaining efficient operation at high switching frequencies.

$$t_r, t_f < 0.05 \times \frac{T}{2} = 83ns \quad (2.9)$$

$$I_{source} = \frac{26nC}{867mA} = 313mA \quad (2.10)$$

2.3.2 Gate Driver

Amplification of the Audio signal occurs at the Half bridge output stage. The half-bridge power stage consists of two power MOSFETs (high-side and low-side) connected in series across the supply rails, with the output taken from their common node. To prevent simultaneous conduction of both devices (a condition known as shoot-through, which causes a low-impedance short circuit across the supply and can destroy the MOSFETs), a short delay – the dead time – is inserted between the turn-off of one device and the turn-on of the other.

Considerations when designing the gate driver:

- Frequency of Triangular wave $\rightarrow 300kHz$
- Comparator output $\rightarrow 0V - 5V$
- Supply voltage to the half bridge output stage $\rightarrow 15V$.
- Gate charge of IRF530 $\rightarrow 26nC$

Gate driver requirements:

- Should be able to handle a switching frequency of $300kHz$
- Should source and sink
- The input should be able to handle $-3V \rightarrow 6V$ addressing voltage spikes induced by parasitic inductance and capacitance.

The gate driver should be able to drive the high and low side Mosfets with two complementary PWM signals inverted with respect to each other. A bootstrap circuit should be implemented to drive the High side mosfet.

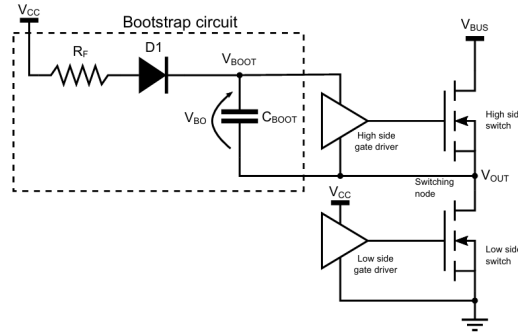


Figure 2.10: Bootstrap circuit

Calculation of bootstrap capacitor value

The bootstrap capacitor should be sized to have enough energy to drive the gate of the high-side MOSFET without being depleted by more than 10%. This bootstrap cap should be at least 10 times greater than the gate capacitance of the high-side FET.

$$C_g = \frac{Q_g}{V_{DD} - V_{\text{Boot Diode}}} \quad (2.11)$$

Where Q_g is the gate capacitance of the Mosfet (IRF530) used.

$$C_g = \frac{26nC}{15 - 0.6} C_g = 1.8nC \quad (2.12)$$

$$C_{boot} > 10C_g \quad (2.13)$$

C_{boot} was chosen to be $100nC$.

The charge to replenish the bootstrap capacitor must come from some larger bypass capacitor, usually the V_{DD} bypass capacitor. As a rule of thumb, this bypass capacitor should be sized to be at least 10 times larger than the bootstrap capacitor so that it is not completely drained during the charging time of the bootstrap capacitor. This allows the bootstrap capacitor to be properly replenished during the charging sequence.

$$C_{VDD} > 10C_{boot} \quad (2.14)$$

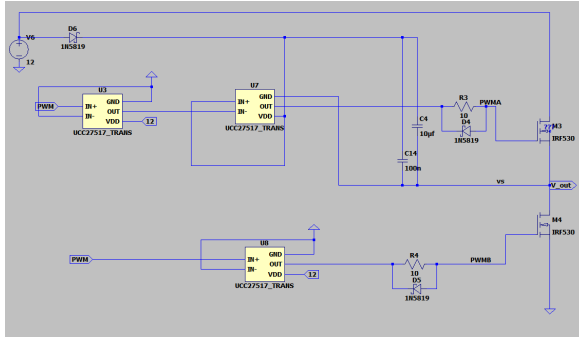
C_{VDD} was chosen to be $1\mu F$.

A Schottky diode (1N5819) was used as the external bootstrap diode.

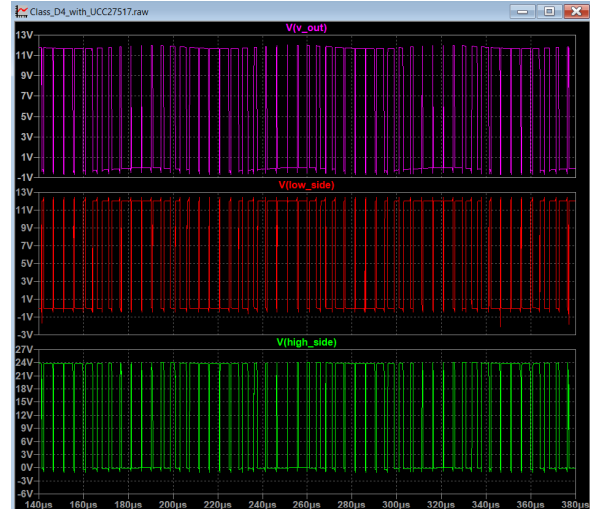
We went through 3 iterations for the design of the Gate driver.

Iteration 1

Implemented using a Low side gate driver UCC27517. Implemented an external Bootstrap circuit.



(a) Iteration 1



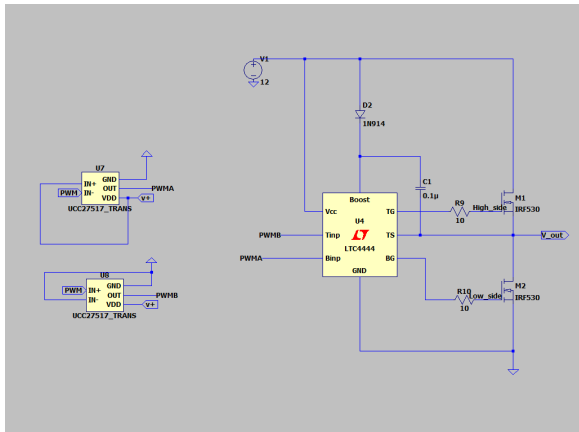
(b) Simulation results of iteration 1

Figure 2.11: Iteration 1 circuit and corresponding simulation results

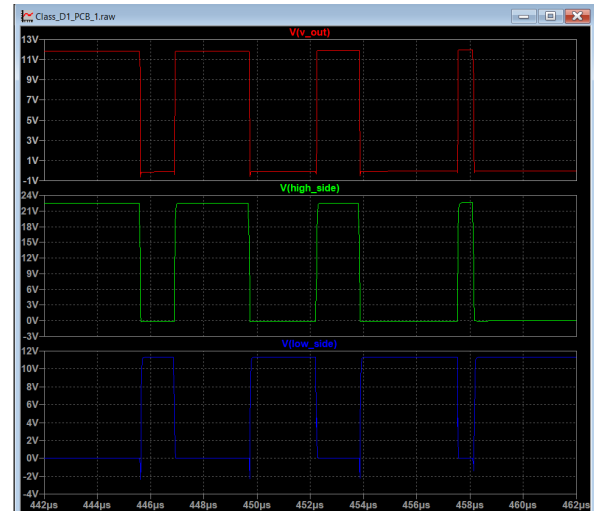
The problem with this circuit was implementing deadtime. Since the duty cycle varies, a simple RC - delay circuit with a fixed delay would not suffice. Adding an asymmetric gate drive network with a diode and resistor also proved futile when implemented on the breadboard. A high current was drawn due to cross conduction.

Iteration 2

Implemented using the LTC4444 high frequency high voltage gate drive. Adaptive shoot-through protection prevents both MOSFETs from conducting simultaneously. The UCC27517 was used as an inverter and pulse shaper to provide clean inputs to the LTC4444.



(a) Iteration 1



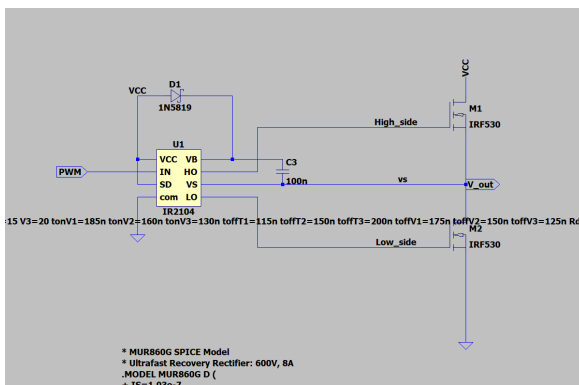
(b) Simulation results of iteration 1

Figure 2.12: Iteration 2 circuit and corresponding simulation results

We planned to implement this circuit on a PCB. After cost and time considerations we moved onto to the third and final iteration.

Iteration 3

Used the IR2104, which implements dead time and internally inverts the pwm signal and produces separate outputs for the High side and Low side Mosfets.



(a) Iteration 3



(b) Simulation results of iteration 3

Figure 2.13: Iteration 2 circuit and corresponding simulation results

Using the IR2104 led us to decrease the triangular wave frequency to 150kHz.

Table 2.2: Gate Driver Comparison: LTC4444 vs IR2104 for Half-Bridge Stage

Parameter	LTC4444	IR2104
Supply Voltage V_{CC}	7.2–13.5 V	10–20 V
Output Configuration	Bootstrap	Bootstrap
Bootstrap Voltage (max)	114 V	600 V offset
Peak Source Current	2.5 A (TG), 3 A (BG)	130 mA
Peak Sink Current	1.2 Ω TG, 0.55 Ω BG pull-down	270 mA
Rise Time (t_r)	8 ns (TG), 6 ns (BG)	680 ns*
Fall Time (t_f)	5 ns (TG), 3 ns (BG)	150 ns*
Propagation Delay (HS)	25 ns	680 ns
Propagation Delay (LS)	19 ns	150 ns
Dead Time	Adaptive shoot-through protection	520 ns fixed
Input Logic	Independent HS/LS inputs	PWM + SD pin
Shoot-through Protection	Adaptive	Built-in
Suitability at 300 kHz	Excellent	Acceptable

The LTC4444 provides substantially higher gate-drive current, lower propagation delay, and significantly faster switching transitions than the IR2104. These characteristics make it highly suitable for high-frequency operation around 300 kHz. The IR2104 offers a simpler implementation with integrated dead-time generation and shoot-through protection, but its larger propagation delays reduce timing accuracy and duty-cycle range at high switching frequencies.

2.4 Filter

2.4.1 Filter Type selection

Table 2.3: Output Filter Type Selection for Class D Amplifier

Parameter	Active Filter	Passive RC	Passive LC
Power Dissipation	Low	High (due to R)	Near-zero (lossless L, C)
Efficiency at Full Power	Poor (op-amp rails needed)	Very poor	Excellent
Output Impedance	Low (op-amp driven)	$= R$ (loads signal)	Very low in passband
Rolloff (2nd order)	-20 dB/decade	-20 dB/decade	-20 dB/decade
Noise Added	Yes (op-amp noise floor)	Thermal noise in R	Near zero
Supply Rails Required	Yes	No	No
Component Count	High	Low	Moderate
PWM Switching Ripple Rejection	Moderate	Poor (limited rolloff)	Excellent
Suitable for >1 W Output	No	No	Yes
Failure Risk	Op-amp breakdown, latch-up	Thermal failure of R	None (no active devices)
Verdict	Not suitable	Not suitable	Correct choice

An active filter cannot survive the rail-to-rail PWM voltage swing at the half-bridge output (12 V, 50 V, or higher depending on supply), as no practical op-amp operates under these conditions. A passive RC filter, while structurally simpler, dissipates substantial power in the series resistor whenever output current flows, making it thermally and efficiency-wise impractical at any meaningful power level. The passive LC filter is the only viable option: inductors and capacitors are ideally lossless, present near-zero impedance in the passband, carry the full output current without dissipation, and provide the steep rolloff required to suppress switching-frequency harmonics while passing the audio band unattenuated.

2.4.2 Cut-off Frequency and Component Selection

In order to obtain constant gain accross all the frequencies ranging 20Hz to 20kHz we chose butterworth 2nd order filter initially. Since it has a maximally flat passband. Initially the Pwm frequency was 350khz however due to the limitations of implementing it on the breadboard we lowered the pwm frequency to 150Khz. Hence we needed much steeper rolloff, So we chose 4th order LC butterworth filter.

Following is the Calculations done for component selection.

Note on R_S mismatch:

Since $R_S = 0.5\Omega$ in the MOSFETs ,not equal to $R_L = 8\Omega$ source/load case.

Cutoff Frequency: 20Khz

Amax : max passband ripple: 3db

Source/Load Ratio (Unmatched R_S , R_L) and prototype values from the table

$$\frac{R_L}{R_S} \approx \frac{8}{0.5} \approx 16$$

$$L_1 = 1.531, \quad C_2 = 1.577$$

$$L_3 = 1.082, \quad C_4 = 0.383$$

Frequency Scaling Factor

$$k_f = \frac{2\pi f_C}{\sqrt{10^{A_{worst}/10} - 1}}$$

$$k_f = 2\pi f_C$$

$$k_f = 2\pi \times f_C = 125,663.706 \text{ rad/s}$$

Impedance Scaling Factor

$$k_m = \frac{R_L}{1} = 8 \Omega$$

Component Scaling Equations

$$C' = \frac{C}{k_m k_f}$$

$$L' = \frac{k_m}{k_f} L$$

Scaled Component Values

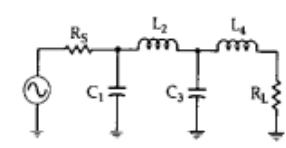
$$C'_1 = \frac{C_1}{k_m k_f} = 1.56 \mu\text{F}$$

$$C'_2 = \frac{C_2}{k_m k_f} = 380 \text{ nF}$$

$$L'_1 = \frac{k_m}{k_f} L_1 = 97.46 \mu\text{H}$$

$$L'_2 = \frac{k_m}{k_f} L_2 = 68.82 \mu\text{H}$$

**Table 3-2A. Butterworth Low-Pass
Prototype Element Values**



n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.111	1.035	1.835		
	1.250	0.849	2.121		
	1.429	0.697	2.439		
	1.667	0.566	2.828		
	2.000	0.448	3.346		
	2.500	0.342	4.095		
	3.333	0.245	5.313		
	5.000	0.156	7.707		
	10.000	0.074	14.814		
	∞	1.414	0.707		
3	0.900	0.808	1.633	1.599	
	0.800	0.844	1.384	1.926	
	0.700	0.915	1.165	2.277	
	0.600	1.023	0.965	2.702	
	0.500	1.181	0.779	3.261	
	0.400	1.425	0.604	4.064	
	0.300	1.838	0.440	5.363	
	0.200	2.009	0.284	7.910	
	0.100	5.167	0.138	15.455	
	∞	1.500	1.333	0.500	
4	1.111	0.466	1.592	1.744	1.469
	1.250	0.366	1.695	1.511	1.811
	1.429	0.325	1.862	1.291	2.175
	1.667	0.269	2.103	1.082	2.613
	2.000	0.218	2.452	0.883	3.187
	2.500	0.169	2.986	0.601	4.009
	3.333	0.124	3.883	0.507	5.338
	5.000	0.080	5.684	0.331	7.940
	10.000	0.039	11.094	0.162	15.642
	∞	1.531	1.577	1.082	0.383
n	R_L/R_S	L_1	C_2	L_3	C_4

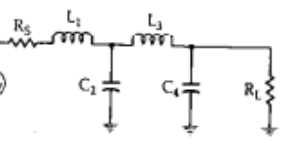
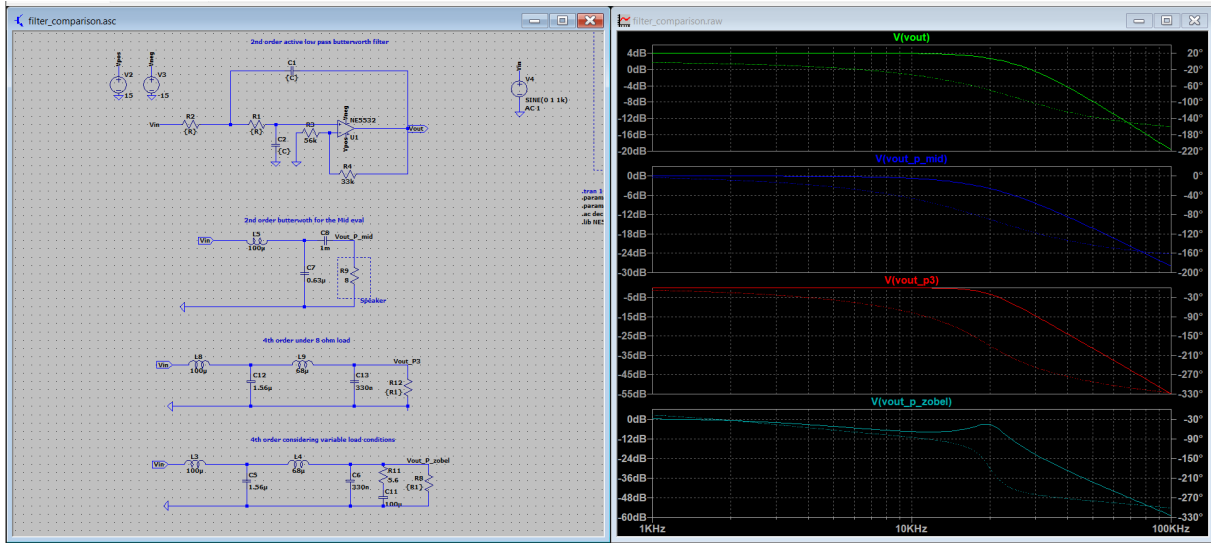


Figure 2.14: Butterworth Low-Pass Prototype Element Values and corresponding LC ladder topology, used as the basis for the unscaled C_1 , C_2 , L_1 , L_2 values above.

Practicle component value

$$C'_2 = 1.56 \mu\text{F}, C'_4 = 330 \text{ nF}, L'_1 = 100 \mu\text{H}, L'_3 = 68 \mu\text{H}.$$

2.4.3 Filter Response and THD Considerations



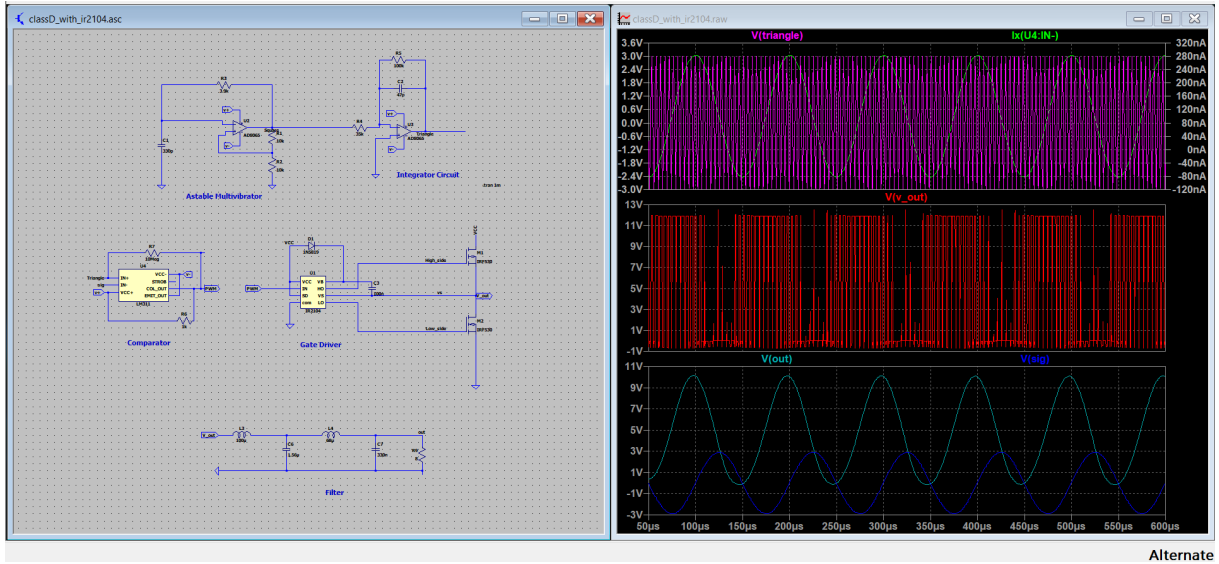
The corresponding AC sweep (1 kHz–100 kHz) magnitude and phase Bode plots for each configuration are shown on the right, illustrating the rolloff and phase behavior differences introduced by load variation and filter order.

LTspice comparison of output filter topologies for the Class D inverter stage:

- (top) 2nd-order active low-pass Butterworth filter (NE5532-based) used as the ideal magnitude/phase reference;
- (upper-middle) 2nd-order passive Butterworth LC filter with 8 Ω load, designed for the mid, a rolloff of 20db can be seen;
- (lower-middle) 4th-order passive LC ladder filter under fixed 8 Ω load This was the design implemented on the final Evaluation has a -3db cutoff at 20Khz and a rolloff of -40db;
- (bottom) 4th-order filter re evaluated under variable/Zobel compensated load conditions. This was designed as a future improvement. The reason being the connected speaker is not purely resistive it has inductance and the actual impedance varies with the frequency therefore a zobel network with $R = 5.6 \Omega$ and $C = 100\mu F$ is added to stabilize the filter response across variable loads.

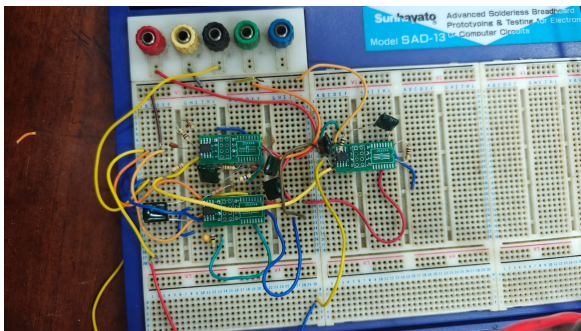
2.5 Simulation Results

2.5.1 End-to-End System Simulation

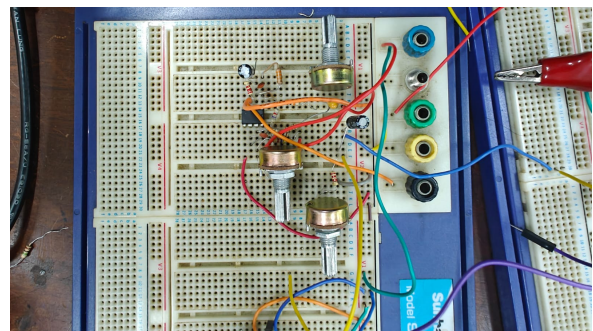


- The first plot shows a triangular wave and the input audio signal which are compared to generate the logic level PWM.
- The second plot shows the PWM output from the gate driver created from the logic PWM from the comparator. This output is used to drive the load, which varies between 0 and 12V.
- This plot shows the amplified audio signal given to the speaker (green) and the input audio (blue). Which clearly shows the signal has been amplified. The output waveform is obtained after the PWM wave is passed through the Filter which removes high frequencies and gives a smooth sine wave.

2.6 Breadboard Implementation



(a) Triangular wave & comparator



(b) Pre amplifier

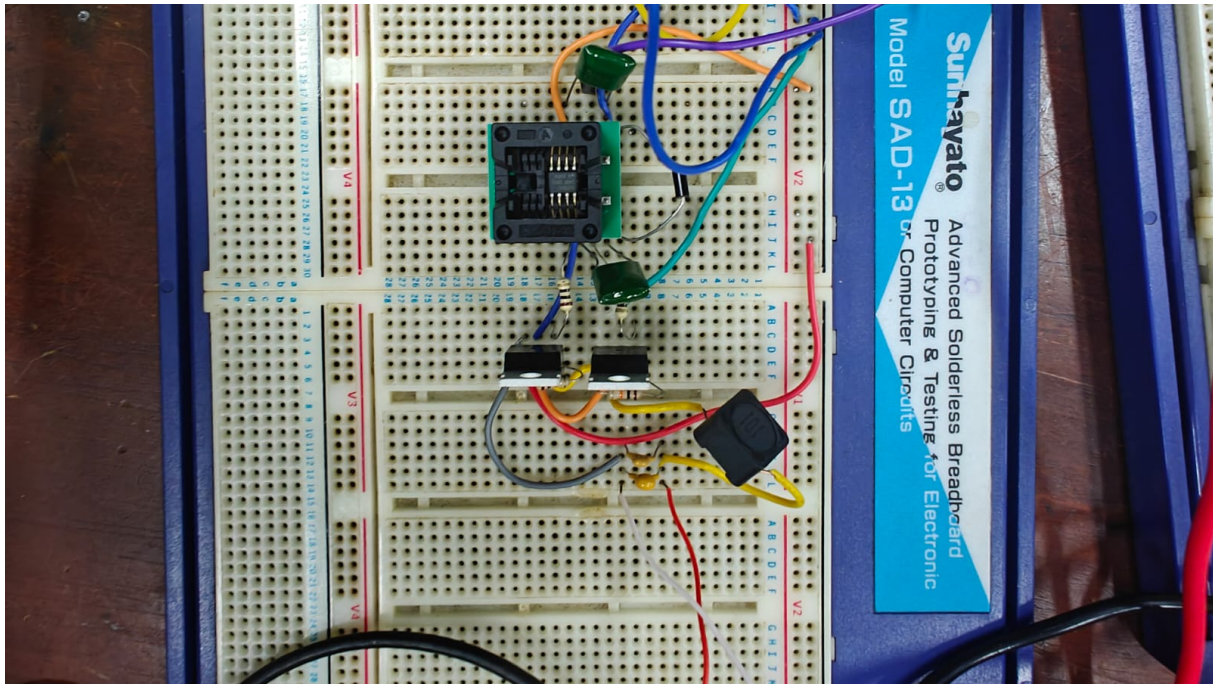
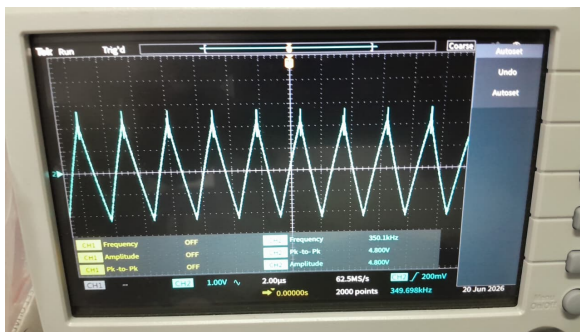
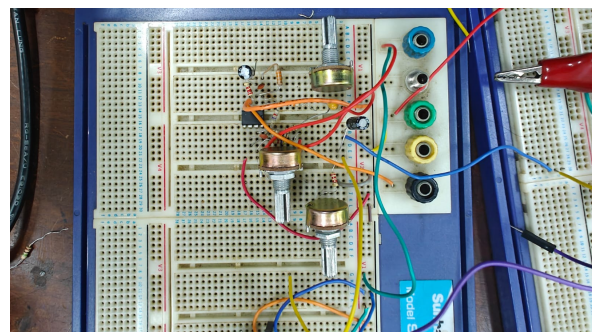


Figure 2.16: Driver and the Filter

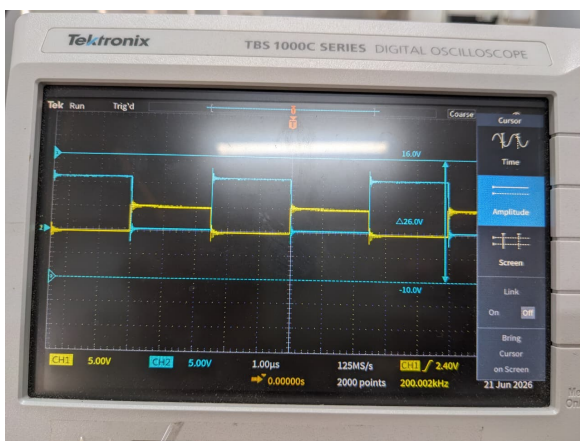
2.6.1 Observations from the oscilloscope



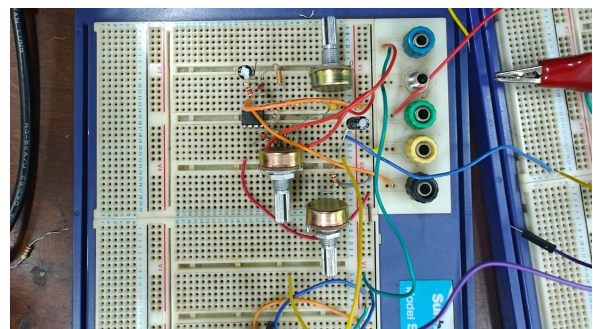
(a) Triangular wave and PWM



(b) Pre amplifier



(a) Gate driver



(b) Filter

2.7 PCB Design

Here we have included the designed PCB. This was done due to the limitations of implementing high frequencies on the breadboard. However we couldn't manufacture the PCB for the final evaluation.

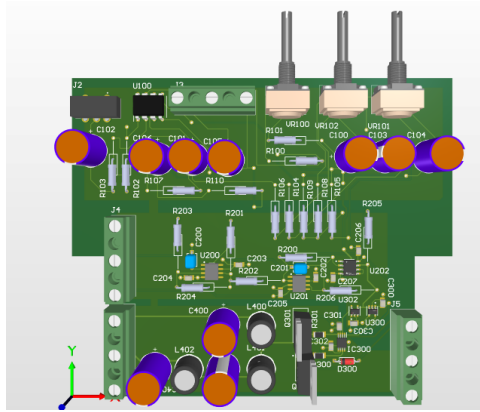
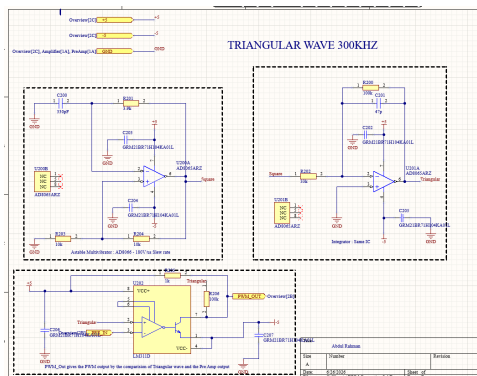
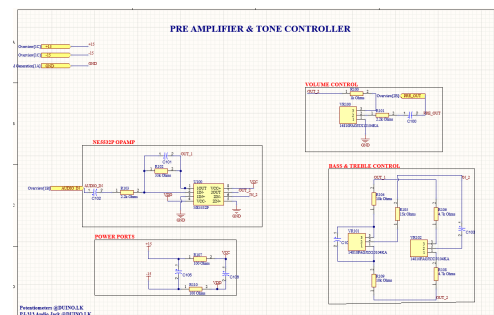


Figure 2.19: PCB

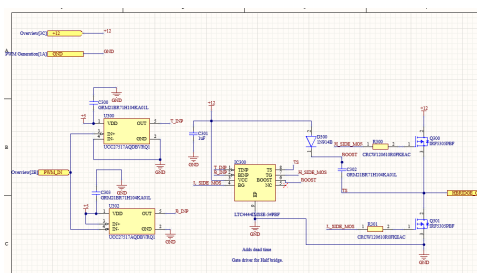
2.7.1 Schematic



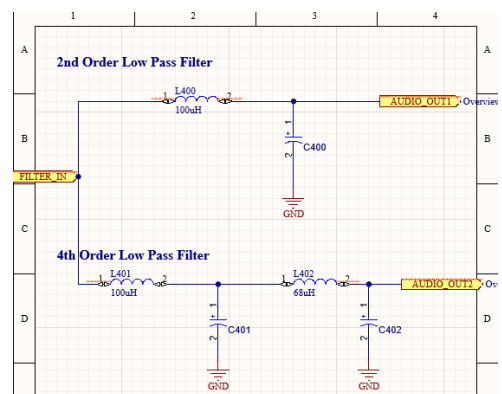
(a) Triangular wave & Comparator



(b) Pre amplifier



(c) Gate driver



(d) Filter

Figure 2.20: Simulation schematics

Conclusion

This project successfully demonstrated the design, simulation, and implementation of a complete Class-D audio amplifier. The system was developed in stages, starting from audio pre-amplification and tone control, followed by PWM generation using a triangular carrier and comparator, and ending with a MOSFET half-bridge output stage and LC filter.

The pre-amplifier provided proper signal conditioning and improved modulation depth for PWM generation. The LM311D comparator improved switching performance compared to earlier designs. The AD8065-based triangular wave generator produced a stable high-frequency carrier suitable for PWM operation, with minor practical noise due to parasitic effects and layout limitations.

In the power stage, the IRF530 was selected for its lower gate charge, enabling better high-frequency switching performance. The IR2104 gate driver provided a reliable solution for the half-bridge, though the switching frequency had to be reduced for stable operation.

The LC output filter successfully reconstructed the audio signal, with the 4th-order Butterworth design giving better attenuation of switching noise and improved audio quality compared to lower-order filters. Simulation and experimental results showed good agreement.

Limitations included breadboard parasitics affecting high-frequency performance, reduced switching frequency compared to the original target, and the absence of PCB fabrication, which limited further optimization.

Future improvements include PCB fabrication with better layout, improved gate driver design with adaptive dead-time control, refinement of the LC filter for real speaker loads, and use of higher-efficiency switching devices such as modern MOSFETs or GaN transistors.

Overall, the project achieved a working Class-D amplifier and provided practical experience in PWM design, power electronics, and mixed-signal system integration.

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